



PAPER ID-310309

Subject Code: BCS055

Roll No:

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BTECH
(SEM V) THEORY EXAMINATION 2024-25
MACHINE LEARNING TECHNIQUES

M.MARKS: 70**TIME: 3 HRS**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A**1. Attempt all questions in brief.****2 x 07 = 14**

Q no.	Question	CO	Level
a.	What constitutes a well-defined learning problem in machine learning? Give an example.	1	K1
b.	How does machine learning differ from data science, and how do they complement each other?	1	K2
c.	Explain the role of the sigmoid function in logistic regression.	2	K2
d.	What is the significance of the decision surface in Support Vector Machines (SVM)?	2	K2
e.	What is overfitting in decision tree learning, and how can it be avoided?	3	K1
f.	Explain the role of the activation function in a perceptron.	4	K2
g.	Describe the role of the reward function in reinforcement learning.	5	K2

SECTION B**2. Attempt any three of the following:****07 x 3 = 07**

Q no.	Question	CO	Level
a.	Describe the steps involved in designing a machine learning system. Illustrate each step with a real-world example.	1	K3
b.	Explore the difference between Simple Linear Regression and Multiple Linear Regression. How does the inclusion of multiple features affect the model?	2	K2
c.	Illustrate the steps of the ID3 algorithm, including how it selects attributes and constructs the tree.	3	K3
d.	Discuss the Self-Organizing Map (SOM) algorithm. How does SOM work for clustering and dimensionality reduction? Explain the differences between SOM and supervised learning models.	4	K2
e.	What is the role of the Q-learning function in reinforcement learning? Explain how it helps the agent learn optimal actions and policies.	5	K3

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

Q no.	Question	CO	Level
a.	Discuss the history of machine learning. Highlight major milestones and advancements in the field.	1	K2
b.	Compare and contrast supervised, unsupervised, and reinforcement learning. Provide examples where each type is applicable.	1	K3

4. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Consider the following weather data to determine whether or not a person will play tennis on a given day based on weather conditions:	2	K3

[illegible]

6. Attempt any <u>one</u> part of the following:		07 x 1 = 07	
Q no.	Question	CO	Level
a.	Derive the mathematical steps involved in backpropagation for training a neural network.	4	K4
b.	Explain the architecture and functioning of Convolutional Neural Networks (CNNs)	4	K4

7. Attempt any <u>one</u> part of the following:		07 x 1 = 07	
Q no.	Question	CO	Level
a.	Explain the concept of Reinforcement Learning (RL). Discuss how it differs from supervised and unsupervised learning.	5	K3
b.	Explain the components of a genetic algorithm. Discuss the role of chromosomes, genes, and the population in the GA cycle.	5	K3